

Exploring Potential of Electromyography-based Avatar Operation Using Residual Muscles of ALS Individuals: Case Study on Avatar DJ Performance

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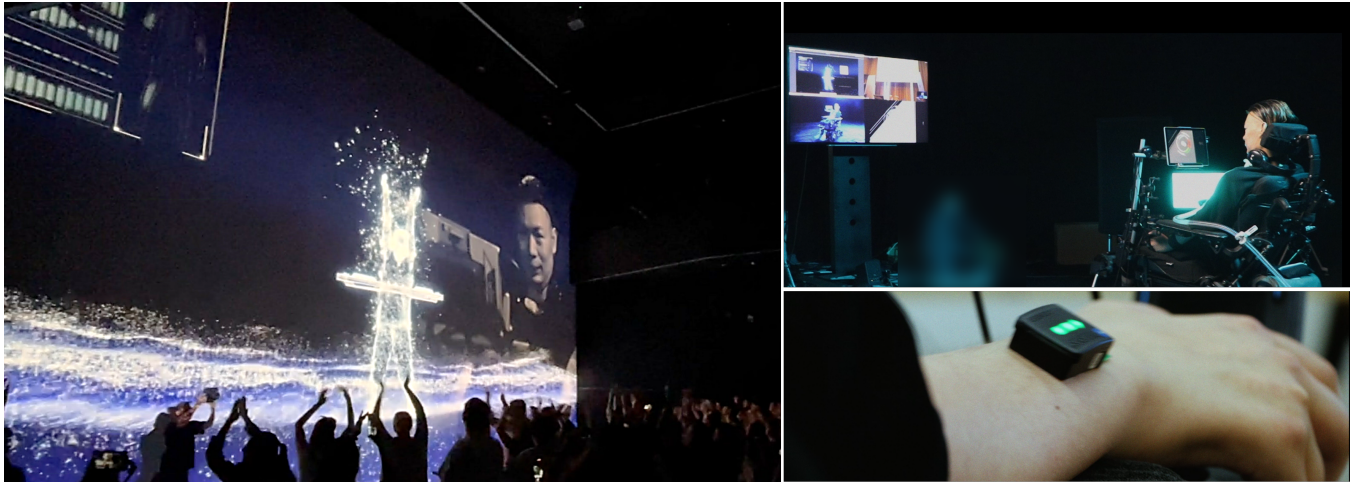


Figure 1: Live performance using our EMG-based avatar communication system at Ars Electronica Festival. Left: DJ performance by a virtual avatar that attracted a large audience. Upper right: an ALS user operating an avatar using the system remotely. Lower right: an example of EMG sensors attached to multiple residual muscles

ABSTRACT

Assistive medical device, such as gastrostomy and ventilation, plays a crucial role in extending the lifespan of individuals with amyotrophic lateral sclerosis (ALS). Additionally, augmentative and alternative communication (AAC) systems are essential for maintaining their ability to communicate. Designing an effective AAC system requires balancing the user's expressive capability with the usability of the user interface. In this study, we focus on not gaze input and electroencephalograph (EEG) signal input commonly used in AAC systems but electromyography (EMG) signal input as a novel modality to augment their embodiment into a virtual space. Since EMG signals can be measured even with slight movements of residual muscles, EMG-based AAC systems provide a potential avenue for communication. We successfully demonstrated the feasibility of our EMG-based avatar communication system by enabling

Masatane Muto, a DJ musician and creator living with ALS, to perform live remotely from Japan to the Ars Electronica Festival 2023 in Linz, Austria using the system. Through this case study and a pilot study, we aimed to explore the potential of EMG-based AAC systems by discussing three perspectives: augmented embodiment, prolonged use, and extensibility into real environments in diverse contexts.

CCS CONCEPTS

• Human-centered computing → User centered design.

KEYWORDS

amyotrophic lateral sclerosis (ALS), electromyography, avatar operation

*Both authors contributed equally to this research.

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1 INTRODUCTION

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease [25, 26, 31]. Individuals with ALS lose the ability to walk, speak, eat, and ultimately breathe without assistive medical devices such as ventilation and gastrostomy due to progressive muscle weakness [14, 20]. Since the goal of ALS care is not only to prolong life but also improve quality of life (QoL) [9], it is important to study the relationship between life-sustaining treatments and QoL.

There have been various studies on the psychological factors affecting the QoL of individuals with ALS after life-prolonging treatments [2, 7, 8, 15, 16]. It has been reported that receiving life-prolonging treatment tends to have a positive effect on the QoL of ALS individuals despite severe physical restrictions [2, 15, 16]. The common requirements from these studies are as follows: 1) the need to be accepting of medical devices and having a positive outlook toward the disease from being adequately informed [2, 15, 16], 2) receiving adequate medical or nursing services after life-prolonging treatment [7, 8], and 3) finding sources of enjoyment, purpose, or meaning in life [7, 8, 16]. To achieve the third common point, it is crucial to research and develop augmentative and alternative communication (AAC) systems that facilitate rich interpersonal relations and performance such as nonverbal expression in a virtual space and playing instruments.

The most common AAC systems are using gaze input and brain-signal input such as electroencephalograph and functional near-infrared spectroscopy [3, 6, 12, 25, 28]. As intellectual functions are preserved even if physical functions including eye movement are completely lost as ALS progresses, the expectation is that brain-computer interfaces (BCIs) using brain signals will become a final input method [18, 19]. However, there are still several challenges that must be overcome for BCIs to be clinically effective [19]. Gaze-input devices are on the market, and many individuals with ALS have used them as text-input and web browsing methods [6, 21, 32]. ALS individuals can speak various languages in their past voice quality by using the combination of gaze-text input and cross-lingual text-to-speech technology [30], which can create a synthesized voice from just a few seconds of voice obtained from past recordings. The synthesized audio created with this technology was used in a live music performance at the ALS Awareness Music Festival "MOVE FES." hosted by WITH ALS on June 18 in Japan, in conjunction with World ALS Day on June 21 [5, 22]. Ongoing research in gaze input has resulted in musical compositions [11] and instrument [29].

Our focus is on electromyography (EMG) signal input as a new modality to augment embodiment of individuals with ALS into a virtual space. The surface EMG sensors can measure the EMG signals even with slight movements of residual muscles. We present a case study featuring DJ performances by an individual with ALS utilizing EMG-based avatar operation through subtle muscle activities in a live event (Figure 1). Additionally, we conduct a pilot study on evaluating our EMG-based AAC system based on a basic physical movement and the DJ performance of the ALS individual in a laboratory environment. The objective of our research is to explore the potential of EMG-based AAC systems, and we discuss three perspectives based on observed evidence from the case study

and the pilot study: augmenting embodiment, prolonged use, and extensibility into real environments in diverse contexts.

2 EMG-BASED AVATAR COMMUNICATION SYSTEM

Our EMG-based avatar communication system is based on residual muscle signals to facilitate rich communication for ALS individuals. Masatane Muto, who is a famous ALS individual and globally active as a DJ musician and creator [1, 5, 10, 23], cooperated with the implementation of the system over approximately half a year. As a first use case, we introduce the operation of a virtual avatar DJ performance using Mr. Muto's residual muscle movements to fulfill his strong desire to once again perform that the way he used to before the progression of ALS. We describe our system in detail in the following subsections.

2.1 Muscle Selection for Avatar Operation

We conducted interviews with Mr. Muto and electromyography measurements to assess his residual muscle functionality. We identified six muscle sites for avatar operation: the left and right splenius muscles involved in the lateral flexion movement of the neck (LNeck, RNeck), left extensor pollicis longus muscle involved in thumb extension (LHand), right extensor indicis muscle involved in index finger extension (RHand), and left and right soleus muscles involved in ankle plantar flexion (LLeg, RLeg).

2.2 Avatar Movement Corresponding to Body Movement

We determined the avatar movements on the basis of how he used to freely move his body when performing in front of an audience before the progression of ALS. The movements of LNeck and RNeck correspond to swaying the avatar's hips to the left and right, creating a rhythmic movement. LHand and RHand correspond to pointing or waving with the left and right hands, respectively. Moving either LLeg or RLeg enables jumping or walking. This design augments slight body movement and enriches avatar communication.

2.3 Hardware

We attached EMG sensors (Delsys, Trigno Wireless System) to the six muscle sites mentioned in the above subsection and measured the EMGs at 1 kHz. The EMG sensors were connected to a PC for EMG processing and sending avatar-movement commands. A tablet-type PC was placed in front of Mr. Muto. He could choose avatar movements corresponding to the movements of their legs and hands by gaze input on the screen shown in Figure 2. This is a function to input EMG signals corresponding only to the selected avatar movement, preventing unintended EMG input. The EMG-processing PC and tablet-type PC for gaze input were both connected to the PC for avatar-movement generation. That is, the gaze input reflects the user's will to move in a certain way, and the EMG signal input reflects the actual movement.

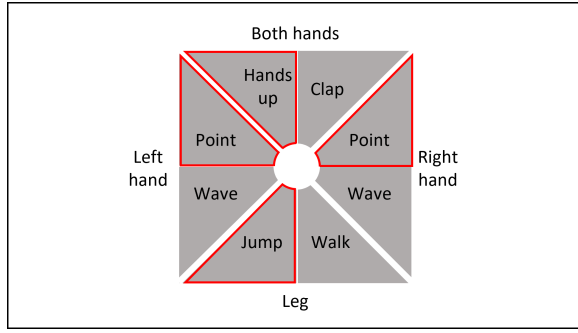


Figure 2: Schematic of screen for selecting avatar movements corresponding to movements of legs and hands by gaze input. Selected avatar movements are highlighted with red border.

2.4 Software

We calculated the root mean square (RMS) every 0.2 s using the EMG data from each muscle site. When the RMS exceeded a specific threshold, a command signal was sent to the PC for avatar-movement generation. The threshold was determined in advance by calculating the RMS during a specific resting period and setting it as a multiple of that constant value. The threshold can be adjusted beforehand according to user preferences but can also be dynamically modified on the basis of the current RMS value. Command information derived from EMG processing and information on avatar-movement selection from the tablet-type PC were consolidated on the avatar-movement-generation PC, resulting in the generation of the corresponding avatar movement, implemented using Unity. We set the duration of avatar movement for each command input to be 3 s on the basis of his operability. If repetitive motor commands are input, the avatar continues its movement for the next 3 s from the last input.

3 LIVE PERFORMANCE USING EMG-BASED AVATAR OPERATION AT ARS ELECTRONICA FESTIVAL

Mr. Muto showcased a live performance using our system on the stage at the Ars Electronica Festival 2023 held in Linz, Austria from September 6th to 10th, 2023, which is the largest annual festival of art, advanced technology, and culture [1, 5, 10, 23]. Figure 1 shows images of the live performance. The avatar on the stage, controlled by the movements of Mr. Muto's six muscle sites at a remote location, performed as the DJ synchronized with his music. Our exhibition made a significant global impact by demonstrating the potential of our system to enable richer forms of communication for ALS individuals.

Mr. Muto said in the interview during rehearsals that "I am very excited because it was a feeling of being able to expand my body, not just supplement the bodily functions I lost due to ALS. I am looking forward to freeing my body after a long time and being able to dance and clap with audiences around the world with my music." He also left the following post-stage message on his blog: "By operating my own avatar through EMG sensors, I extended my physical presence, energizing the audience at the venue."

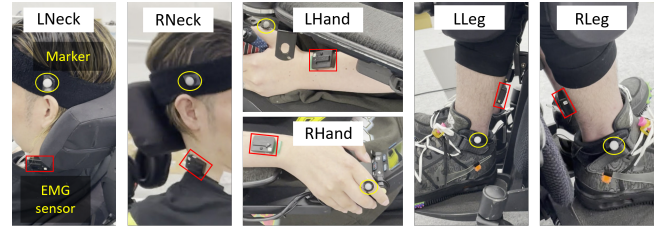


Figure 3: Positions of motion-capture markers and EMG sensors.

4 PILOT STUDY IN LABORATORY ENVIRONMENT

We conducted a pilot study in a laboratory environment involving Mr. Muto to understand two main aspects: the characteristics of his residual body movements and the differences in the comfort of avatar control with varied avatar-movement durations. The former was assessed by measuring EMG and movement of the residual muscles using EMG sensors and a motion-capture system. The latter was discussed through interviews about how he used his body. This basic investigation provides useful information to support Mr. Muto's comments in the previous section about his physical sensations and to evaluate the balance between his expressive capabilities and the usability of the UI during the live performance.

4.1 Ethics

This experiment was reviewed and approved by the local ethics committee (approval number: 2023-07, date of approval: 4th October 2023). Mr. Muto was informed of the experimental protocol and signed a consent form before the experiment. He agreed to publicly disclose his name and personal images in this paper.

4.2 Apparatus and Analysis

The setup for our EMG-based avatar system remained consistent with that presented in the previous section. Avatar movements were assigned as follows: swaying hips using the neck (LNeck and RNeck), hands up with the hands (LHand and RHand), and jumping with the legs (LLeg and RLeg). We recorded the time-series changes in RMS data calculated from EMG signals of the target muscles and the timing of avatar-movement commands.

To measure the movements of each target body part, an infrared reflective marker from the motion capture system (Vicon, Vicon Motion Capture Systems) was attached to each part (Figure 3) and eleven motion capture cameras tracked their 3D positions at 100Hz, synchronized with our EMG-based avatar communication system. Each 3D position data point was differenced from the initial position and the norm was calculated, representing the distance from the initial position.

4.3 Procedure

The experimental procedure was as follows. We conducted open interviews with Mr. Muto at each step. He took sufficient breaks between the steps.

- Step 1: EMG calibration: He rested for 3 s in the sitting position and the threshold was adjusted to ensure comfortable operation.
- Step 2: He voluntarily moved the instructed body part for 10 s in accordance with the experimenter's signal. Between movements, he took a 10-s rest. He was instructed to move within a comfortable range and was guided to repeat applying force if the strength weakened. The order of the instructed parts was randomized.
- Step 3: He performed step 2 by changing the avatar-movement duration for 1, 2, and 3 s in response to a single command input.
- Step 4: He performed the DJ performance with the same music during the live performance. The avatar-movement duration was set to 3 s on the basis of the interview.

We investigated the characteristics of his residual body movements using the quantitative time-series changes, interview results from step 2, the differences in user comfort related to the duration of avatar movements on the basis of the interview from step 3.

4.4 Results

4.4.1 Time-Series Changes in EMG and Corresponding Motion. Figure 4 shows the time-series changes in the marker position as the distance from the initial position (blue line), RMS calculated from EMG signals (red line), and avatar command (green line) for each body part in step 2 of the procedure. The yellow highlights indicate the body parts instructed to move. Examining the highlighted areas in the figure reveals that clear EMG signals were observed even for subtle movements ranging from a few millimeters to centimeters. Examining the non-highlighted areas revealed unintended muscle activity that was not instructed to move, particularly in the neck and left leg, resulting in unintended avatar movements. Mr. Muto expressed concern, stating, "The occurrence of unintended avatar movements is worrisome".

4.4.2 Feedback on Comfort Related to Duration of Avatar Movements. We obtained feedback on the differences in comfort when controlling the avatar by varying the duration of avatar movements with a single command input to 1, 2, and 3 s in step 3 of the procedure. Mr. Muto expressed a preference for a 3-s duration, stating, "In this type of avatar movement, setting a 3-second duration increased comfort by conserving the frequency of my movements. With a 1-second duration, muscle fatigue quickly occurred, making it challenging to sustain communication for a long period." We then asked about any discomfort or misalignment in operation due to the long duration of avatar movements. He responded, "Since I had learned the characteristics of the avatar's movements in advance, there was no discomfort. It could be overcome through practice." We also asked if he felt a sense of moving his own body. He mentioned, "Even with a low frequency of movement, if I could achieve the planned avatar movements as intended, I could feel like I was moving my own body." We explored whether he wanted to change the duration depending on the form of avatar operation. The response was, "Generally, it's better to get used to the 3-second setting. If more precise movements are required, 1 second might be better, but it will be for professionals."

5 DISCUSSIONS

Through a series of practices with an ALS individual using our EMG-based avatar communication system, we discuss the potential of this system from three perspectives: augmenting embodiment, the points for prolonged communication use, and extensibility for use in diverse contexts.

5.1 Augmenting Embodiment

According to Mr. Muto's positive feedback about his subjective embodiment regarding the global live performance, our EMG-based system was effective in augmenting embodied sense in the body. The pilot study also found that he could maintain a sense of moving his own body despite inputs using minimal body movements and a 3-s continuation of avatar motion per input. His answer regarding the sense of moving his own body, as discussed in 4.4.2, implies that the key to augmenting embodiment through virtual avatars is to voluntarily move the avatars and achieve the pre-planned avatar actions as intended, even if the corresponding avatar movements are not perfectly aligned with the body movements. Previous studies have shown that a strong correlation and followability of body movements in virtual and real space is necessary for the generation of embodiment [13, 27], but we assume that different conditions exist for users with limited body movements in this study. We speculate that the reason for the difference from previous findings is due to different expectations of an avatar for ALS individuals; able-bodied individuals expect an avatar to result in strong similarity to their movements, whereas ALS individuals expect their unmoving bodies to move via avatars.

5.2 Prolonged Communication Use

Sustaining physical communication for longer periods is important to improve the quality of life for ALS individuals. Mr. Muto's DJ performance lasted 5 min, which seems a very long time to continue with intermittent physical movements. According to his response about comfort related to the duration of avatar movements in 4.4.2, a longer continuation of the avatar motion per EMG input was comfortable for him in reducing muscle fatigue, suggesting that the effects of muscle fatigue need to be taken into account for prolonged communication with our system. Individuals with severe physical disabilities such as ALS have diminished muscle strength and endurance [26, 31], so they prefer to conserve movements while maintaining intended operations. Therefore, input methods that involve long muscle contraction or continuous input based on continuous EMG signals are less favorable. In previous human-interface research, the approach for more realistic outputs in response to high-speed inputs seems to be increasingly important [17, 24]. However, this study involving an ALS participant suggests a contrasting perspective. For such users, achieving planned avatar movements with minimal input appears more crucial than responsiveness, because they can continue to communicate longer by reducing muscle fatigue. Previous studies have reported that AAC systems are used flexibly according to the progress of ALS [4], and the present findings can serve as a guideline for balancing the expressive capabilities of severe ALS individuals and the operability of the UI of the EMG-based AAC system.

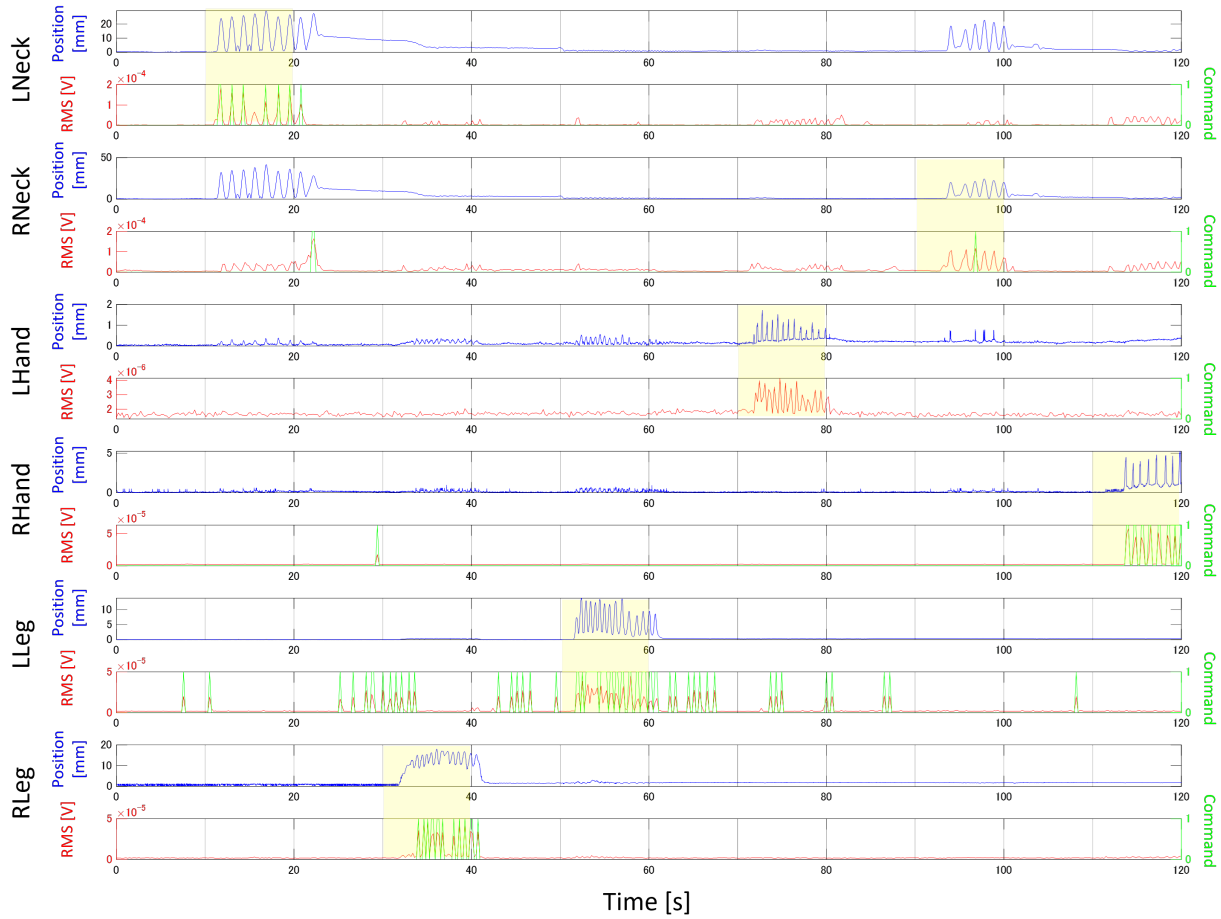


Figure 4: Time-series changes in marker position as distance from initial position (blue line), RMS (red line), and avatar command (green line) for each body part. Yellow highlights indicate body parts instructed to move.

5.3 Extensibility to Spread Into Real Environments in Diverse Contexts

Through a series of practices, several challenges should be addressed to spread this system into real environments in diverse use cases and user physical conditions. The case in the live performance at a world-class arts festival is one example. During the performance, increased muscle tension throughout his body due to Mr. Muto's tension and excitement heightened the possibility of unintended inputs. Unintended inputs from other sites and his negative feedback, as confirmed in the laboratory experiment, suggested that this element could diminish embodiment. We attempted to solve this from two perspectives: technical-system improvement and third-person assistance. The former was an automatic calibration based on the resting state immediately before the performance and the selection of avatar movements using gaze input, which was not a perfect solution and may not be sufficient in terms of actual use. The latter was that a third person, who was one of the authors, dynamically adjusted the thresholds of avatar commands to assist him, which led to the success of the show. However, the

supporter may be required to have an understanding of the user's intentions and expertise in motor control. We found it difficult for non-researchers with poor user understanding and knowledge to determine the timing and amount of threshold modifications. These findings indicated that both technological advances to improve operability and enhancement of supporters to assist the technologies are necessary to widely spread EMG-based AAC systems, which supports the previous statement that other aspects than operability should also be discussed in AAC research in HCI community [3].

6 LIMITATIONS AND FUTURE WORK

Although this study was only a case study of one person with ALS and one use case, we obtained knowledge that can be utilized as a basis for further studies with more users and for application to other use cases. We plan to investigate the adaptability of the system to various aspects, such as the degree of symptoms and motivation to use a new interface. In addition, we will consider a variety of cases, such as customer service via robots, communication between avatars in the metaverse, and use in eSports.

7 CONCLUSION

We developed an EMG-based avatar communication system that is based on residual muscle signals to augment embodiment of ALS individuals. A famous DJ performer with ALS showcased a live performance using our system at the worldwide festival and our exhibition made a significant global impact by demonstrating the potential of our system to enable richer forms of communication for ALS individuals. We conducted a pilot study to deeply understand his comments about his physical sensations and to evaluate the balance between his expressive capabilities and the usability of the UI during the live performance. The results and discussions suggested the potential of this system from three perspectives: augmenting embodiment, the points for prolonged communication use, and extensibility for use in diverse contexts. Findings unique to those with physical disabilities were found; the key to augmenting embodiment through virtual avatars is to voluntarily move the avatars and achieve the pre-planned avatar actions as intended, even if the corresponding avatar movements are not perfectly aligned with the body movements.

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